

Cloud Computing Fundamentals

Cloud computing delivers computing services over the internet including servers, storage, databases, networking, and software.

The three main service models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

Major cloud providers include Amazon Web Services, Microsoft Azure, and Google Cloud Platform, each offering hundreds of services.

Benefits include reduced IT costs, scalability, business continuity, collaboration efficiency, and flexibility of work practices.

Cloud Architecture and Deployment

Cloud deployment models include public cloud, private cloud, hybrid cloud, and multi-cloud strategies.

Microservices architecture breaks applications into small, independent services that communicate via APIs.

Containerization with Docker and orchestration with Kubernetes have become the standard for cloud-native applications.

Serverless computing allows developers to build applications without managing infrastructure. AWS Lambda and Azure Functions are popular serverless platforms.